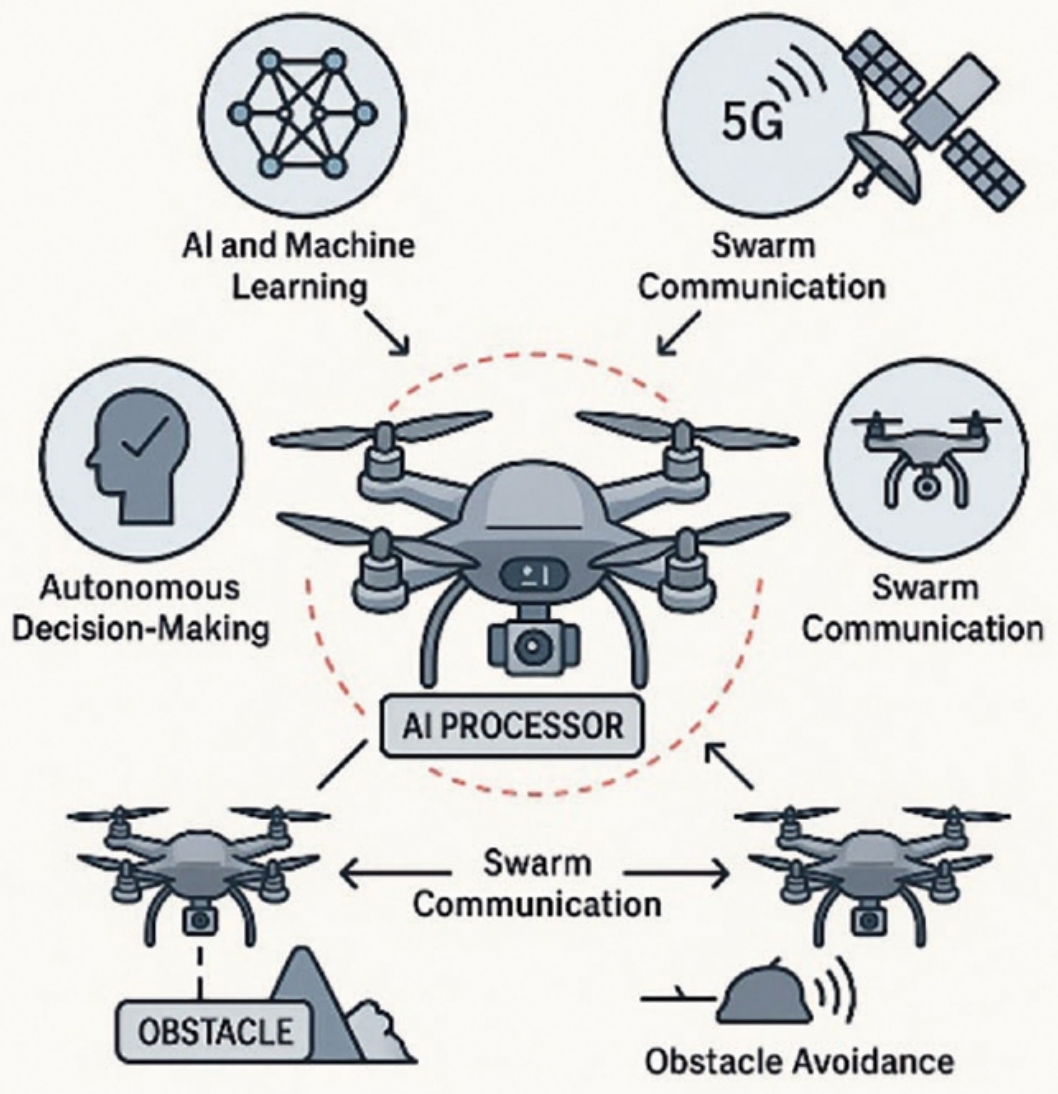



**Figure 1.7** Fifth-generation drone full autonomy, AI, and swarm technology. ⏎

## 1.5 Summary

The chapter provides a comprehensive understanding of the evolution, structure, and impact of unmanned aerial vehicles (UAVs), tracing their transformation from manually operated systems to highly intelligent, autonomous platforms. It explores the technological foundations that have shaped drone development, covering advancements in aerodynamics, electronics, navigation, and artificial intelligence, while outlining their generational progression from early military prototypes to modern AI-